



Unit 1 Safety and Matter

Intensified Chemistry Essential Knowledge and Guided Notes

Read Textbook: CK Chemistry Chapter 1 - Introduction to Chemistry and Chapter 2 - Matter and Change: <https://flexbooks.ck12.org/cbook/ck-12-chemistry-flexbook-2.0/>

Watch Helpful Videos and Practice: Posted on CANVAS

Highlight Essential Knowledge: Use a different color marker or highlighter.

Safety:

The chemistry lab is an inherently dangerous place, with fire hazards, dangerous chemicals and procedures that if done incorrectly can be risky. No one wants to have an accident in the lab, so it is important to follow all lab safety rules. While there are specific safety rules for each experience and experiment in lab this year, the 10 most important to remember are summarized below, but also read the article that is linked here to view details and important information: <https://tinyurl.com/WLBig10SafetyRules>

1. Follow instructions and ask questions if you are unsure.
2. Know the location and use of safety equipment. Be sure you know the hazards of any chemicals used. Every chemical has a Safety Data Sheet (SDS) that should be consulted prior to working with it in the lab.
3. Dress for the lab. The proper clothing and PPE (personal protective equipment like goggles and aprons) are important for protecting yourself.
4. Don't eat or drink in the laboratory. That includes taking water to your lab station.
5. Don't play mad scientist in the laboratory (or ever!).
6. Don't taste or sniff chemicals.
7. Dispose of waste properly. Not everything can go untreated in the trash or sink.
8. Know what to do with lab accidents. The most important first step is telling the teacher.
9. Leave experiments at the lab.
10. Don't experiment on yourself.

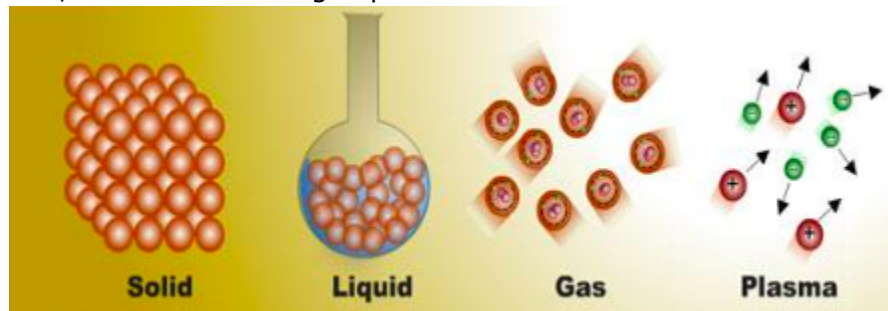
The full safety contract is found posted on CANVAS.

Green chemistry is the design of chemical products and processes that reduce or eliminate the use or generation of hazardous substances. Green chemistry applies across the life cycle of a chemical product, including its design, manufacture, use, and ultimate disposal. Using the principles of Green Chemistry allows for a safer classroom environment.

Matter:

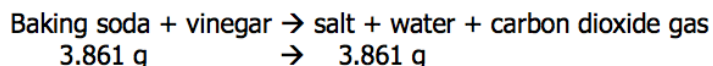
1. **Matter** is the stuff that makes up everything. Matter is anything that has **mass** and **volume**.
2. Mass is the amount of matter in a substance. The **International System of Units (SI)** basic measure of mass is the kilogram (kg), although for smaller units the gram (g) is used. Mass is measured on a device known as a balance or scale.
3. **Volume** is a measure of the amount of space that a substance takes up. The basic SI unit for volume is the cubic meter (m^3), but for smaller volumes may be measured in cm^3 . Liquid volumes are often measured in liters (L) or milliliters (mL).
 - a. Liquid volumes are often measured in graduated cylinders in chemistry
 - b. Gas volumes depend on the shape of the container as gases fill up the entire space occupied.
 - c. Solid volumes can be measured using dimensions and geometric formulas, but irregularly shaped solids are often measured using **water displacement**.

4. A **pure substance** is matter that has a uniform and definite composition. All samples of substances, sometimes called pure substances, have identical properties. A pure substance can be classified by its physical and chemical properties.
5. **Physical properties** refer to the condition or quality of a substance that can be observed or measured without changing the substance's composition. Important physical properties are density, conductivity, melting point, boiling point, malleability (the ability to be molded into shapes), and ductility (the ability to be drawn into wires).
6. Physical properties can be **extensive** or **intensive**.
 - a. Extensive physical properties depend on the amount of the substance, such as the mass of a piece of iron or the volume of liquid water.
 - b. Intensive properties only depend on the type of substance and not the amount such as the conductivity of a copper wire.
7. An important physical property is the **phase or state of matter**: solid, liquid gas or plasma.
 - a. **Solids** have a definite shape and volume. The particles vibrate around a fixed point.
 - b. **Liquids** have a definite volume, but take on the shape of the container. The particles can move around each other, but are still attracted to each other.
 - c. **Gases** do not have a definite volume or shape. The particles have little attraction to each other. Gases are highly compressible.
 - d. **Plasma** is similar to a gas. Like a gas, plasma consists of particles of matter that can pull apart and spread out, so it lacks a fixed volume and a fixed shape. Unlike a gas, plasma can conduct electricity and respond to a magnetic field. That's because plasma consists of electrically charged particles called ions, instead of uncharged particles such as atoms or molecules.



8. A **chemical property** describes the ability of a substance to undergo a specific chemical change. A chemical property of iron is that it is capable of combining with oxygen to form iron oxide, the chemical name of rust. The more general term for rusting and other similar processes is corrosion. Other terms that are commonly used in descriptions of chemical changes are burn, rot, explode, decompose, and ferment. Chemical properties are very useful in identifying substances. However, unlike physical properties, chemical properties can only be observed as the substance is in the process of being changed into a different substance.
9. **Physical and Chemical Changes:** There are several differences between a physical and chemical change in matter or **substances**.
 - a. A physical change in a substance doesn't change what the substance is, but just some of the physical attributes of a substance. An example of a physical change would be tearing paper or boiling water.
 - b. In a chemical change where there is a chemical reaction, a new substance is formed and energy is either given off or absorbed. An example of a chemical change would be iron rusting or water decomposing to form hydrogen and oxygen gases.

10. **The Law of Conservation of Mass or Matter** states that matter cannot be created or destroyed in a chemical reaction. Conservation of matter is represented in balanced chemical equations.



11. **Pure substances** can be classified into two categories: Elements and Compounds.
- An **element** is the simplest form of matter that has a unique set of properties.
 - It is a pure substance that cannot be broken down into different types of substances.
 - As of 2022, there are **118 known elements**. Ninety eight of the elements are naturally occurring on earth, while the other 20 elements have been made synthetically in a lab.
 - The names of all the elements can be found on the **periodic table**. Elements have names, but are also represented by symbols. **Symbols** may be 1 or 2 letters. The first letter of an element is always capitalized and the second letter if needed is lowercase. Examples of symbols are H(Hydrogen), He (Helium), S (Sulfur) and Na (Sodium, based on its Latin name natrium, since "S" is already used by sulfur)
 - Most elements exist as **atoms** in nature, but seven elements exist as **diatomic molecules** in which two of the same type of atom chemically combine. Oxygen exists as a diatomic molecule for example. Its chemical formula in nature is O_2 . The "2" in O_2 is called a subscript and indicates that 2 oxygen atoms are chemically bonded.
 - Along with oxygen, the elements: hydrogen, fluorine, chlorine, bromine, iodine, and nitrogen also exist in nature as diatomic molecules.
 - A **compound** is a pure substance that contains two or more elements chemically combined in a fixed proportion.
 - The elements carbon and hydrogen combine to form many different compounds. One of the simplest is called methane, in which there are always four times as many hydrogen particles as carbon particles. Methane is a pure substance because it always has the same composition. However, it is not an element because it can be broken down into simpler substances – carbon and hydrogen.
 - Compounds are represented by **chemical formulas** of two or more element symbols and consist of two or more elements chemically bonded. The chemical formula of methane is CH_4 .
 - Compounds exist as **molecules** (2 or more atoms chemically bonded) or a **formula unit** (positive and negative ions held together in a fixed ratio by electrostatic attraction)
 - Subscripts** represent the relative number of each type of atom in a molecule or formula unit.
 - Coefficients** are numbers found in front of chemical symbols or formulas to indicate how many of each atom or molecule are present. Parentheses indicate the number of groups of atoms in a formula and are used for "polyatomic ions" which stay together as a group. We will learn about this more later.

Examples: $\text{Ca}(\text{NO}_3)_2 = \text{Ca NO}_3 \text{NO}_3 = 1 \text{ Ca } 2 \text{ N } 6 \text{ O}$

Symbol or Formula	Number and type of atoms	With Coefficient	Number and type of atoms
LiNO_3	1 Li 1 N 3 O (5 total atoms)	3 LiNO_3	3 Li 3 N 9 O (15 total atoms)
CO_2	1 C 2 O (3 total atoms)	4 CO_2	4 C 8 O (12 total atoms)
$(\text{NH}_4)_3\text{PO}_4$	3 N 12 H 1 P 4 O (20 total atoms)	2 $(\text{NH}_4)_3\text{PO}_4$	6 N 24 H 2 P 8 O (40 total atoms)

vi. Compounds have different properties than the elements from which they are composed.

12. A **mixture** is a combination of two or more substances in any proportion. They are NOT pure substances.
- This is different from a compound, which consists of substances in fixed proportions.
 - The substances in a mixture also do not combine chemically to form a new substance, as they do in a compound. Instead, they just intermingle and keep their original properties. Lemonade is a mixture because it doesn't have fixed proportions of ingredients. It could have more or less lemon juice, for example, or more or less sugar, and it would still be lemonade.
 - Mixtures may be **homogeneous** with a very uniform composition throughout or **heterogeneous** where each sample of the mixture may yield a slightly different composition (like sampling Trail Mix).
 - Homogeneous mixtures are called "solutions." A **solution** is a homogeneous mixture with tiny particles. The particles are too small to see and also too small to settle or be filtered out of the mixture
13. Mixtures can be separated based on the **physical properties** of its components. Three common methods of separation are distillation, filtration and chromatography.
- Distillation** separates mixtures by differences in boiling point
 - Filtration** separates mixtures of undissolved solids from liquids by particle size.
 - Chromatography** separates mixtures based on attractive forces between particles in solutions.

U1 Lesson #1 Guided Notes - Safety and Green Chemistry

Video Notes: Crash Course Chemistry - Lab Techniques & Safety
(It is also found embedded in the lesson if you are absent)

- Why should you not use glassware that you know is not totally clean?
- Why should long hair be "UP"?
- What are some do's and don'ts about lab attire/clothing?

4. **TRUE/FALSE:** Glasses count as eye protection.
5. Where is the eye wash station (look around in the classroom)?
6. What should you do if your eyes feel funny?
7. What should you do if you think you have been exposed to a hazardous chemical?
8. **TRUE/FALSE:** Eating and drinking in the lab is a good idea.
9. On the Haz-Mat diamond give 2 examples of what a 4 rating can do.
10. What is an SDS for? Why would you use it?
11. What is the fume hood for? Where is it (look around in the classroom)?
12. What is the proper way to smell a substance?_____
13. Why is it important to COMMIT when pouring a liquid?
14. How do you deal with broken glass?
15. **TRUE/FALSE:** It is fine to dump all lab materials down the sink.
16. Where is the shower and the fire extinguisher (look around the room)?
17. What can you do to be as SAFE as possible in a lab class?
18. Why do you need to be careful about disposing of all chemicals in the same container?

Green Chemistry Notes:

19. What is Green Chemistry?
20. List the 12 principles of Green Chemistry.
 1. Prevent _____
 2. Maximize _____
 3. Design _____ chemical syntheses
 4. Design _____ chemicals and products
 5. Use safer _____ and reaction _____
 6. Increase _____ efficiency

7. Use _____ feedstocks
8. Avoid chemical _____
9. Use _____, not stoichiometric reagents
10. Design chemicals and products to _____ after use
11. Analyze in real time to prevent _____
12. Minimize the potential for _____

U1 Lesson #2 Guided Notes - Defining Chemistry & Matter

1. Chemistry is
 - the branch of science that deals with the identification of the substances of which _____ is composed;
 - the investigation of their _____ and the ways in which they interact, combine, and change;
 - and the use of these processes to form _____.
2. How is matter defined in chemistry?
3. What are the only things that are NOT matter?

4. Define Mass and Volume:

	Definition	Units
Mass		
Volume		

5. Define Pure Substances:

Definition:	
Examples:	

6. **TRUE/FALSE:** All samples of a pure substance are always identical and have the same properties. These properties may be physical or chemical properties.

7. Define Physical Properties:

Definition:	
Examples:	

8. Define Extensive and Intensive Properties

	Definition	Examples
Extensive		
Intensive		

9. **TRUE/FALSE** The phase of matter is a physical property.

10. What are the four phases of matter?

_____ have a definite shape and volume. The particles vibrate around a fixed point.

_____ have a definite volume, but take on the shape of the container. The particles can move around each other, but are still attracted to each other.

_____ do not have a definite volume or shape. The particles have little attraction to each other. Gases are highly compressible.

_____ is similar to a gas. Like a gas, plasma consists of particles of matter that can pull apart and spread out, so it lacks a fixed volume and a fixed shape. Unlike a gas, plasma can conduct electricity and respond to a magnetic field. That's because plasma consists of electrically charged particles called ions, instead of uncharged particles such as atoms or molecules.

11. Draw a representation of each phase of matter below using dots to represent the particles and their relative placement and arrangement by referring to the picture in the slides:

Solid	Liquid	Gas	Plasma

12. Define Chemical Properties:

Definition:	
Example:	

13. How are chemical properties unlike physical properties?

14. Physical vs. Chemical Changes:

Type	Definition
Physical Change	
Chemical Change	

15. What type of changes are phase changes? _____

16. Label the Endothermic Phase Changes - Adding Energy (Enter!)

Solid → Liquid (s) → (l) = _____

Liquid → Gas (l) → (g) = _____

Solid → Gas (s) → (g) = _____

17. Label the Exothermic Phase Changes - Taking Energy Away (Exit!)

Liquid → solid (l) → (s) = _____

Gas → Liquid (g) → (l) = _____

Gas → Solid (g) → (s) = _____

18. Chemical vs. Physical Changes Example Vocabulary:

Type	Example Words
Physical Change	
Chemical Change	

19. List the 4 signs of a chemical change:

a.
b.
c.
d.

20. What is the law of conservation of mass or matter? Explain an example:

U1 Lesson #3 Guided Notes -Classifying Matter: Element, Compound and Mixture

1. What is a pure substance?
2. Pure substances can be classified into two categories: _____ and _____.
3. An _____ is the simplest form of matter that has a unique set of properties. It is a pure substance that cannot be broken down into different types of substances. As of 2022, there are _____ known elements. _____ elements are naturally occurring on earth, while the other _____ elements have been made synthetically in the lab.
4. What is the Periodic Table?

5. Elements are represented by _____. They may be **one or two** letters. The first letter is _____. The second letter is _____.

6. Why is a second letter sometimes needed?

7. Most elements exist as _____ in nature, but seven elements exist as _____ in which two of the same type of atom chemically combine. _____ are particles in which more than one atom is bonded chemically to another.

8. Draw the diagram from the slides of Helium and Nitrogen in the boxes below:

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9. Which is "diatomic" helium or nitrogen? ____ Why?

10. List the formulas of the seven diatomic elements:

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11. What is a subscript?

12. Define Compound:

Definition:	
Example:	

13. Chemical Formulas:

- Compounds are represented by _____ of two or more element symbols and consist of two or more elements chemically bonded.
- Compounds exist as _____ (2 or more atoms chemically bonded) or a _____ (positive and negative ions held together in a fixed ratio by electrostatic attraction)
- _____ represent the relative number of each type of atom in a molecule or formula unit.

14. Sketch the following representations of compounds:

Compound (CO ₂)	Formula Unit (NaCl)

15. _____ are numbers found in front of chemical symbols or formulas to indicate how many of each atom or molecule are present. _____ indicate number of groups of atoms in a formula and are used for "polyatomic ions" which stay together as a group. We will learn about this more later.

Example: $\text{Ca}(\text{NO}_3)_2 = \text{Ca NO}_3 \text{ NO}_3 = 1 \text{ Ca } 2 \text{ N } 6 \text{ O}$

More Examples:

16. **Label** $2 \text{ C}_8\text{H}_{18}$

17. **TRUE/FALSE** Compounds have the same properties as the elements that make them up.

18. Define Mixture:

Definition:	
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19. How is a mixture different from a compound?

20. **TRUE/FALSE** Mixtures are NOT pure substance.

21. Define Homogeneous and Heterogeneous mixtures

Type	Definition	Example
Homogeneous		
Heterogeneous		

22. Homogeneous mixtures are also called _____.

23. Mixtures can be separated based on the _____ of its components.

Three common methods of separation are distillation, filtration and chromatography.

- _____ separates mixtures by differences in boiling point
- _____ separates mixtures of undissolved solids from liquids by particle size.
- _____ separates mixtures based on attractive forces between particles in solutions.